Week 5 - Lab

Exercise 5-1: Software setup: **deno**

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Dev

Exercise 5-1: Software setup: deno

https://docs.deno.com/

1. Open a **Windows Powershell** terminal
2. Run: `irm https://deno.land/install.ps1 | iex`
3. Close the **Windows Powershell** terminal
4. Open a new **Windows Powershell** terminal
5. test with `deno -v`

```
> deno -v
deno 2.9.6
```

Deno Runtime

GET STARTED

- Welcome to Deno
- Installation
- Setup your environment

GUIDES

- Running code
- Dependency management
- Web development
- JSX and React
- HTTP Server

GET STARTED

Installation

Deno is a single binary executable with no external dependencies. It runs on macOS, Linux, and Windows, on both x64 and arm64 architectures.

Linux | macOS | **Windows**

Using PowerShell (Windows):

```
irm https://deno.land/install.ps1 | iex
```

Exercise 5-2: Software setup: Celbridge v1.4

Download Celbridge **v1.4** <<<<

- (improved for Typescript)

https://celbridge.org/download/

Celbridge

https://celbridge.org/download/

Home Forum ↗ Learn ↗ Download

DOWNLOAD

Get Celbridge

Available for Windows and macOS.

Download for Windows
v1.4.0 (147 MB) ↗

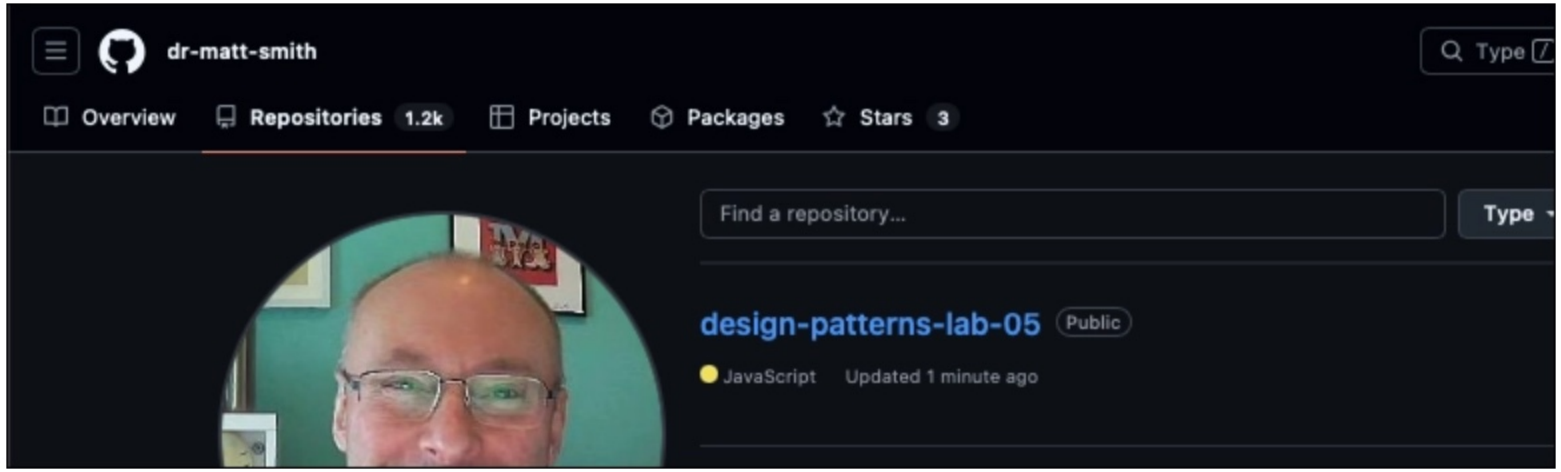
Download for Mac
v1.4.0 (138 MB) ↗



Exercise 5-3: Download the Lab 5 Github project

Download and unzip the GitHub for week 05:

https://github.com/dr-matt-smith/design-patterns-lab-05



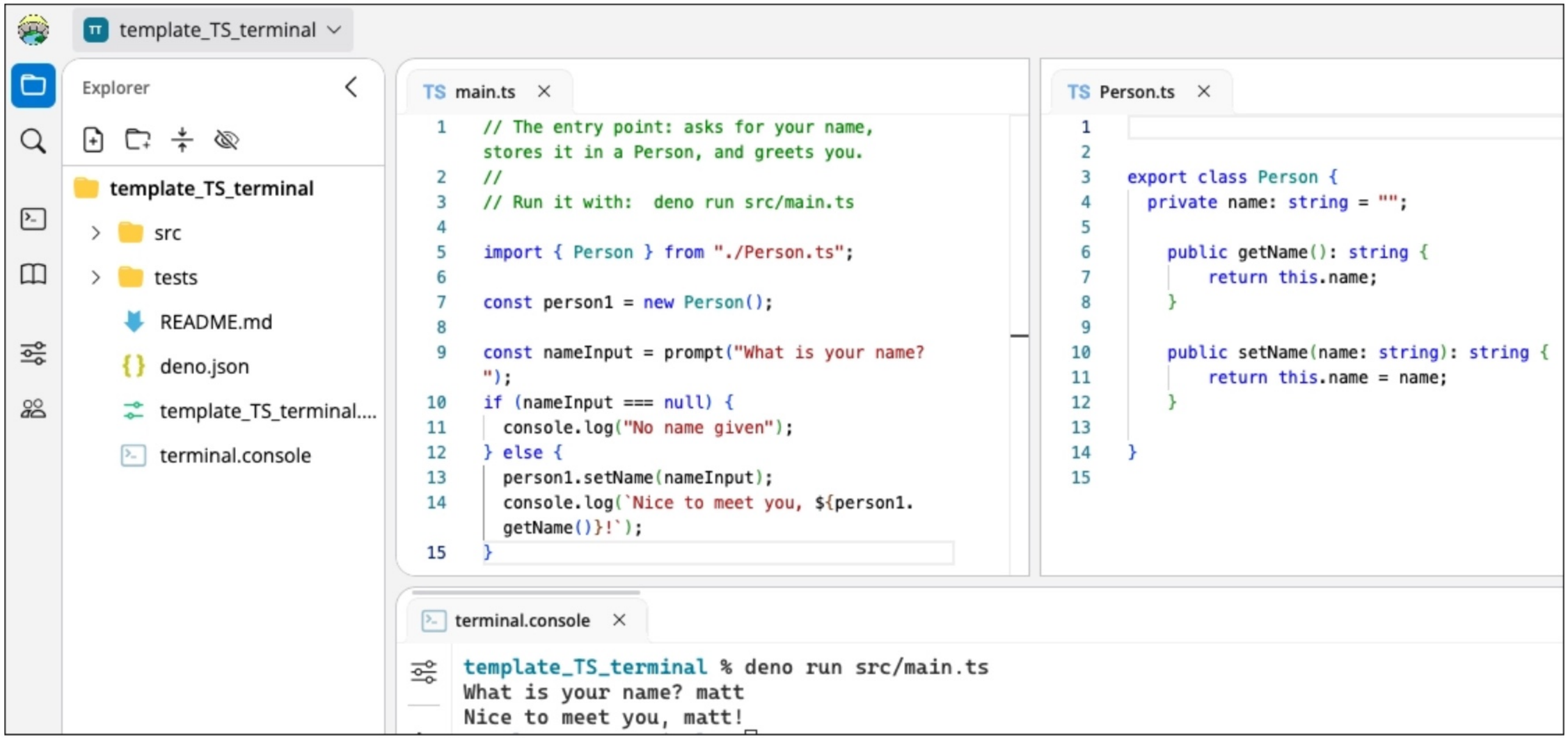
Exercise 5-3: Hello World (terminal)

- Create a copy of Celbridge project `template_TS_terminal`
- look through the code
- update for `firstName` and `surname` properties
  update `main.ts` to input both names
- add a few more tests

You should be happy with running and testing a TypeScript project with deno

- and declaring and using simple classes in TypeScript

Dev



http://127.0.0.1:51595/project/images/hello_matt_terminal.webp

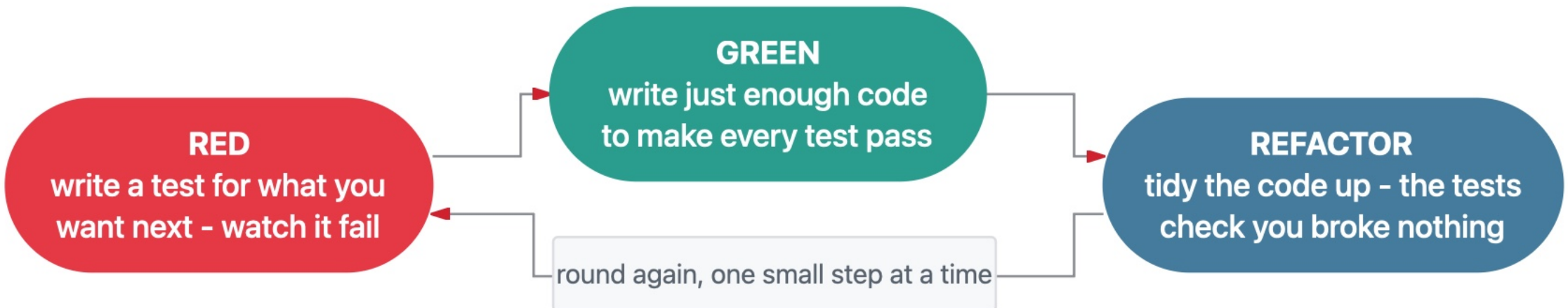
Test-driven development

Try writing a test first

- for `Person.set/getSurname()`

get your RED errors

then write code to pass the test ...



Exercise 5-5: Hello World (web)

- Create a copy of Celbridge project `template_TS_web`
- look through the code
- update for `firstName` and `surname` properties
  update `main.ts` to input both names
- add a few more tests

You should be happy with running and testing a TypeScript web project

try adding a bulleted list to your web project

```
// needs a <ul> in the HTML code with ID #planets
const list = document.querySelector("#planets");

const PLANETS: string[] = ["Mercury", "Venus", "Earth", "Mars"];

for (const planet of PLANETS) {
  const item = document.createElement("li");
  item.textContent = planet;

  // add an <li> element to the list with current planet name
  list.appendChild(item);
}
```

- `for ... of` - not `for ... in`
- `.length` - no brackets

#Exercise 5-6: explopre TS Observer pattern resources

Explore the contents of folder

`observer_pattern_TS`

